

XYZ Frizy Firstyaji – 5024221073

Laporan Environment Variable and Set-UID Program Lab

[illegible]

Task 2: Passing Environment Variables from Parent Process to Child Process.

```
ek@j0yy: ~/Downloads/Labsetup
$ cd Downloads/Labsetup
$ sudo nano nyrintrm.c
$ ./a.out
SESSION_MANAGER=local:/j0yy-8/tmp:/ICE-unix/2811,unix:/j0yy:/tmp:/ICE-unix/2811
QT_ACCESSIBILITY=1
COLORTERM=truecolor
XDG_CONFIG_DIRS=/etc/xdg/xdg-ubuntu:/etc/xdg
SSH_AGENT_LAUNCHER=gnome-keyring
XDG_MENU_PREFIX=gnome-
GNOME_DESKTOP_SESSION_ID=this-is-deprecated
GNOME_SHELL_SESSION_MODE=ubuntu
SSH_AUTH_SOCK=/run/user/1000/ssh-agent/ssh
XNOFILES=gnome-bus
DESKTOP_SESSION=ubuntu
GTK_MODULES=gail:atk-bridge
PANGO_PATH=/home/ek31/Downloads/Labsetup
LOGNAME=ek31
XDG_SESSION_DESKTOP=ubuntu
XDG_SESSION_TYPE=wayland
SYSTEMD_EXEC_PID=2835
XAUTHORITY=/run/user/1000/.mutter-Xwaylandauth.BQWCH3
HOME=/home/ek31
USER=root
XDG_CONFIG_PHASE=1
LANG=en_US.UTF-8
LS_COLORS=rs=0:di=0:34:ln=0:136:ln=0:36:rh=0:36:pi=0:33:so=0:135:do=0:135:bd=0:43:cd=0:43:or=0:40:31:di=0:ml=0:37:41:sg=0:34:ca=0:41:tw=0:42:cm=0:34:42:st=0:37:44:ex=0:32:*.tar=0:31:*.tgz=0:31:*.arc=0:31:*.arj=0:31:*.taz=0:31:*.lha=0:31:*.lz=0:31:*.lzh=0:31:*.lma=0:31:*.tlz=0:31:*.ttx=0:31:*.txz=0:31:*.tzo=0:31:*.zip=0:31:*.z=0:31:*.dmg=0:31:*.gz=0:31:*.lrz=0:31:*.lzo=0:31:*.swf=0:31:*.tar.gz=0:31:*.tar.bz2=0:31:*.bz2=0:31:*.bz=0:31:*.bz1=0:31:*.bz3=0:31:*.bz3c=0:31:*.bz3i=0:31:*.bz3l=0:31:*.bz3s=0:31:*.bz3t=0:31:*.bz3u=0:31:*.bz3v=0:31:*.bz3w=0:31:*.bz3x=0:31:*.bz3y=0:31:*.bz3z=0:31:*.bz3a=0:31:*.bz3b=0:31:*.bz3c=0:31:*.bz3d=0:31:*.bz3e=0:31:*.bz3f=0:31:*.bz3g=0:31:*.bz3h=0:31:*.bz3i=0:31:*.bz3j=0:31:*.bz3k=0:31:*.bz3l=0:31:*.bz3m=0:31:*.bz3n=0:31:*.bz3o=0:31:*.bz3p=0:31:*.bz3q=0:31:*.bz3r=0:31:*.bz3s=0:31:*.bz3t=0:31:*.bz3u=0:31:*.bz3v=0:31:*.bz3w=0:31:*.bz3x=0:31:*.bz3y=0:31:*.bz3z=0:31:*.bz3a=0:31:*.bz3b=0:31:*.bz3c=0:31:*.bz3d=0:31:*.bz3e=0:31:*.bz3f=0:31:*.bz3g=0:31:*.bz3h=0:31:*.bz3i=0:31:*.bz3j=0:31:*.bz3k=0:31:*.bz3l=0:31:*.bz3m=0:31:*.bz3n=0:31:*.bz3o=0:31:*.bz3p=0:31:*.bz3q=0:31:*.bz3r=0:31:*.bz3s=0:31:*.bz3t=0:31:*.bz3u=0:31:*.bz3v=0:31:*.bz3w=0:31:*.bz3x=0:31:*.bz3y=0:31:*.bz3z=0:31:*.bz3a=0:31:*.bz3b=0:31:*.bz3c=0:31:*.bz3d=0:31:*.bz3e=0:31:*.bz3f=0:31:*.bz3g=0:31:*.bz3h=0:31:*.bz3i=0:31:*.bz3j=0:31:*.bz3k=0:31:*.bz3l=0:31:*.bz3m=0:31:*.bz3n=0:31:*.bz3o=0:31:*.bz3p=0:31:*.bz3q=0:31:*.bz3r=0:31:*.bz3s=0:31:*.bz3t=0:31:*.bz3u=0:31:*.bz3v=0:31:*.bz3w=0:31:*.bz3x=0:31:*.bz3y=0:31:*.bz3z=0:31:*.bz3a=0:31:*.bz3b=0:31:*.bz3c=0:31:*.bz3d=0:31:*.bz3e=0:31:*.bz3f=0:31:*.bz3g=0:31:*.bz3h=0:31:*.bz3i=0:31:*.bz3j=0:31:*.bz3k=0:31:*.bz3l=0:31:*.bz3m=0:31:*.bz3n=0:31:*.bz3o=0:31:*.bz3p=0:31:*.bz3q=0:31:*.bz3r=0:31:*.bz3s=0:31:*.bz3t=0:31:*.bz3u=0:31:*.bz3v=0:31:*.bz3w=0:31:*.bz3x=0:31:*.bz3y=0:31:*.bz3z=0:31:*.bz3a=0:31:*.bz3b=0:31:*.bz3c=0:31:*.bz3d=0:31:*.bz3e=0:31:*.bz3f=0:31:*.bz3g=0:31:*.bz3h=0:31:*.bz3i=0:31:*.bz3j=0:31:*.bz3k=0:31:*.bz3l=0:31:*.bz3m=0:31:*.bz3n=0:31:*.bz3o=0:31:*.bz3p=0:31:*.bz3q=0:31:*.bz3r=0:31:*.bz3s=0:31:*.bz3t=0:31:*.bz3u=0:31:*.bz3v=0:31:*.bz3w=0:31:*.bz3x=0:31:*.bz3y=0:31:*.bz3z=0:31:*.bz3a=0:31:*.bz3b=0:31:*.bz3c=0:31:*.bz3d=0:31:*.bz3e=0:31:*.bz3f=0:31:*.bz3g=0:31:*.bz3h=0:31:*.bz3i=0:31:*.bz3j=0:31:*.bz3k=0:31:*.bz3l=0:31:*.bz3m=0:31:*.bz3n=0:31:*.bz3o=0:31:*.bz3p=0:31:*.bz3q=0:31:*.bz3r=0:31:*.bz3s=0:31:*.bz3t=0:31:*.bz3u=0:31:*.bz3v=0:31:*.bz3w=0:31:*.bz3x=0:31:*.bz3y=0:31:*.bz3z=0:31:*.bz3a=0:31:*.bz3b=0:31:*.bz3c=0:31:*.bz3d=0:31:*.bz3e=0:31:*.bz3f=0:31:*.bz3g=0:31:*.bz3h=0:31:*.bz3i=0:31:*.bz3j=0:31:*.bz3k=0:31:*.bz3l=0:31:*.bz3m=0:31:*.bz3n=0:31:*.bz3o=0:31:*.bz3p=0:31:*.bz3q=0:31:*.bz3r=0:31:*.bz3s=0:31:*.bz3t=0:31:*.bz3u=0:31:*.bz3v=0:31:*.bz3w=0:31:*.bz3x=0:31:*.bz3y=0:31:*.bz3z=0:31:*.bz3a=0:31:*.bz3b=0:31:*.bz3c=0:31:*.bz3d=0:31:*.bz3e=0:31:*.bz3f=0:31:*.bz3g=0:31:*.bz3h=0:31:*.bz3i=0:31:*.bz3j=0:31:*.bz3k=0:31:*.bz3l=0:31:*.bz3m=0:31:*.bz3n=0:31:*.bz3o=0:31:*.bz3p=0:31:*.bz3q=0:31:*.bz3r=0:31:*.bz3s=0:31:*.bz3t=0:31:*.bz3u=0:31:*.bz3v=0:31:*.bz3w=0:31:*.bz3x=0:31:*.bz3y=0:31:*.bz3z=0:31:*.bz3a=0:31:*.bz3b=0:31:*.bz3c=0:31:*.bz3d=0:31:*.bz3e=0:31:*.bz3f=0:31:*.bz3g=0:31:*.bz3h=0:31:*.bz3i=0:31:*.bz3j=0:31:*.bz3k=0:31:*.bz3l=0:31:*.bz3m=0:31:*.bz3n=0:31:*.bz3o=0:31:*.bz3p=0:31:*.bz3q=0:31:*.bz3r=0:31:*.bz3s=0:31:*.bz3t=0:31:*.bz3u=0:31:*.bz3v=0:31:*.bz3w=0:31:*.bz3x=0:31:*.bz3y=0:31:*.bz3z=0:31:*.bz3a=0:31:*.bz3b=0:31:*.bz3c=0:31:*.bz3d=0:31:*.bz3e=0:31:*.bz3f=0:31:*.bz3g=0:31:*.bz3h=0:31:*.bz3i=0:31:*.bz3j=0:31:*.bz3k=0:31:*.bz3l=0:31:*.bz3m=0:31:*.bz3n=0:31:*.bz3o=0:31:*.bz3p=0:31:*.bz3q=0:31:*.bz3r=0:31:*.bz3s=0:31:*.bz3t=0:31:*.bz3u=0:31:*.bz3v=0:31:*.bz3w=0:31:*.bz3x=0:31:*.bz3y=0:31:*.bz3z=0:31:*.bz3a=0:31:*.bz3b=0:31:*.bz3c=0:31:*.bz3d=0:31:*.bz3e=0:31:*.bz3f=0:31:*.bz3g=0:31:*.bz3h=0:31:*.bz3i=0:31:*.bz3j=0:31:*.bz3k=0:31:*.bz3l=0:31:*.bz3m=0:31:*.bz3n=0:31:*.bz3o=0:31:*.bz3p=0:31:*.bz3q=0:31:*.bz3r=0:31:*.bz3s=0:31:*.bz3t=0:31:*.bz3u=0:31:*.bz3v=0:31:*.bz3w=0:31:*.bz3x=0:31:*.bz3y=0:31:*.bz3z=0:31:*.bz3a=0:31:*.bz3b=0:31:*.bz3c=0:31:*.bz3d=0:31:*.bz3e=0:31:*.bz3f=0:31:*.bz3g=0:31:*.bz3h=0:31:*.bz3i=0:31:*.bz3j=0:31:*.bz3k=0:31:*.bz3l=0:31:*.bz3m=0:31:*.bz3n=0:31:*.bz3o=0:31:*.bz3p=0:31:*.bz3q=0:31:*.bz3r=0:31:*.bz3s=0:31:*.bz3t=0:31:*.bz3u=0:31:*.bz3v=0:31:*.bz3w=0:31:*.bz
```

Ketika **fork()** dipanggil, system membuat Salinan proses **parent** untuk **child** process, termasuk **environment variables** yang dimiliki **parent**. Fungsi **printenv()** dijalankan pada **parent** dan **child**, hasilnya identic, hal ini terjadi dikarenakan **child process** mewarisi process **parent** yang sama “**printenv()**”. Env variable pada kedua proses adalah sama.

The image shows a dual-pane terminal window with two active tabs: 'child_emvxt' and 'parent_emvxt'. Both tabs show a shell script with various environment variables and commands. The 'child_emvxt' script includes variables like SHELL, SESSION_MANAGER, QT_ACCESSIBILITY, and commands like XDG_CONFIG_DIRS, SSH_AGENT_LAUNCHER, XDG_MENU_PREFIX, GNOME_DESKTOP_SESSION_ID, GNOME_SHELL_SESSION_MODE, SSH_AUTH_SOCK, XMODIFIERS, DESKTOP_SESSION, GTK_MODULES, PKD, LOGNAME, XDG_SESSION_DESKTOP, XDG_SESSION_TYPE, SYSTEMD_EXEC_PID, AUTHORITY, HOME, USERNAME, TM_CONFIG_PHASE, LANG, LS_COLORS, WAYLAND_DISPLAY, GNOME_TERMINAL_SCREEN, GNOME_SETUP_DISPLAY, LESSCLOSE, XDG_SESSION_CLASS, TERM, LESSOPEN, USER, GNOME_TERMINAL_SERVICE, DISPLAY, SHLV_L, QT_TK_MODULE, XDG_RUNTIME_DIR, XDG_DATA_DIRS, PATH, and DBUS_SESSION_BUS_ADDRESS. The 'parent_emvxt' script is identical to the 'child_emvxt' script. The terminal window has a top bar with system status icons and a bottom bar with window management controls.

Task 3: Environment Variables and `execve()`.

[illegible]

Environment variables diperlakukan ketika program menjalankan proses baru menggunakan **execve()**. **execve()** berbeda dengan **fork()**, **execve()** memungkinkan program menentukan environment variables yang diberikan ke program baru. Ketika parameter **execve()** berisi **NULL**, maka program tidak bisa menerima environment variable apapun. Hal ini berkaitan dengan hal pewarisan environment variable yang tidak otomatis diberikan, maka dari itu environment harus diberikan secara eksplisit melalui parameter fungsi.

Task 4: Environment Variables and system().

[illegible]

system() menjalankan cmd shell seperti **/bin/sh -c command**, environment variables akan mendapatkan warisan dari proses yang memanggilnya, hal ini mirip dengan parent process dengan menjalankan command melalui shell sehingga environment variables tetaop diwariskan dari parent process.

Task 5: Environment Variable and Set-UID Programs.

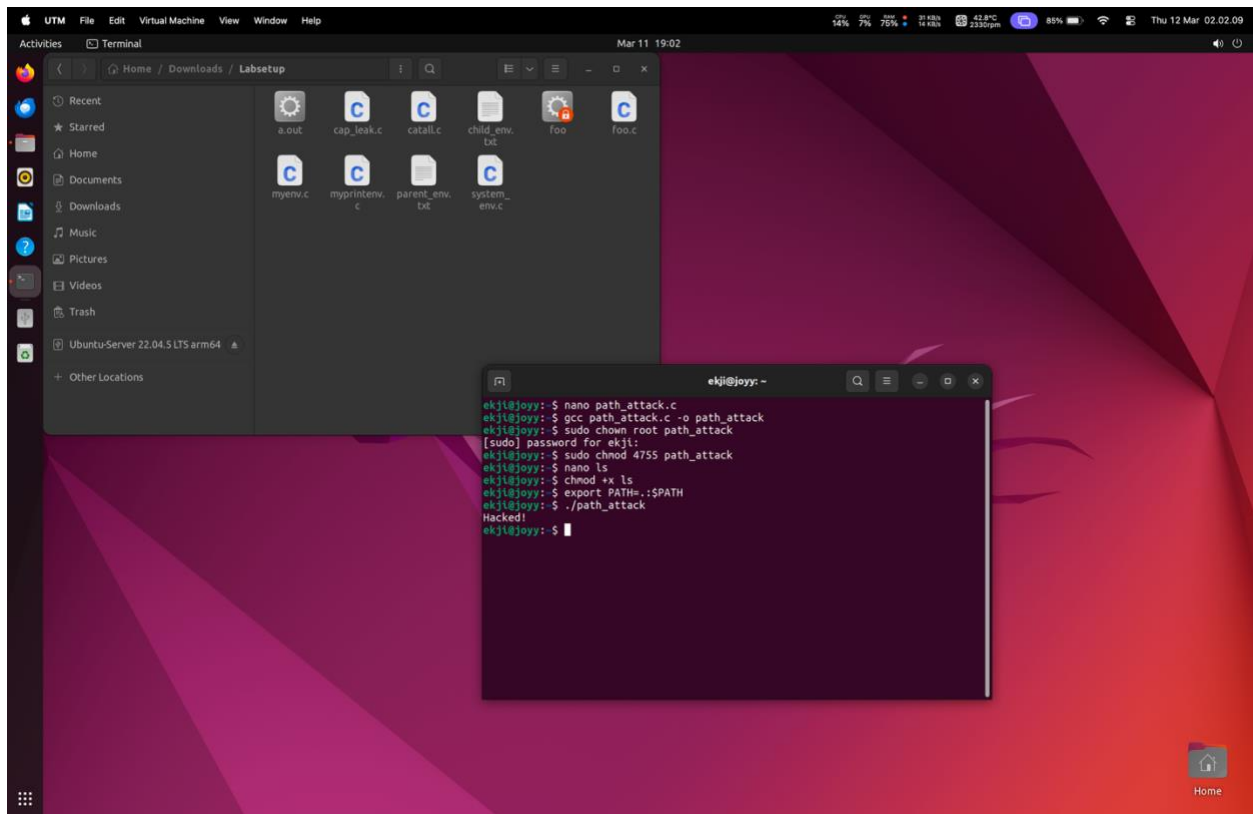
[illegible]

Pada task ini, dibuat program **Set-UID** dengan langkah:

```
sudo chown root foo -sudo chmod 4755 foo
```

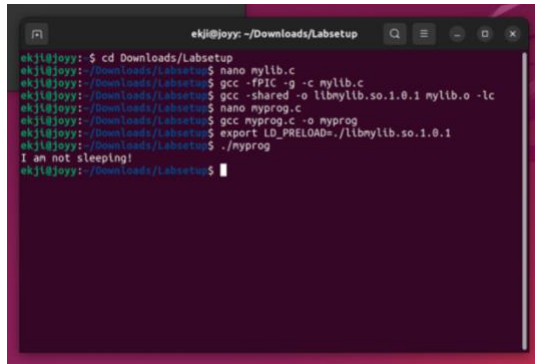
Bit 4755 mengaktifkan **Set-UID Bit**, supaya Ketika program dijalankan oleh user biasa, program berjalan dengan **effective UID** milik owner file, sama dengan root. Ketika program **Set-UID** dijalankan, OS membersihkan atau mengabaikan beberapa environment variable yang berbahaya, karena dapat digunakan untuk melakukan privilege escalation. Program **Set-UID** berjalan dengan hak akses pemilik file yang dibatasi oleh OS untuk keamanan.

Task 6: The PATH Environment Variable and Set-UID Programs.



Pada task ini dilakukan percobaan **PATH ATTACK**, Ketika Program **Set-UID** memanggil command **system("ls")**, pada user directory, terbuatlah file ls palsu dan variable **PATH** diubah menjadi **export PATH=./\$PATH**, dikarenakan saat ini directory berada di ".", system akan menjalankan program **ls** palsu milik user. Kemudian program **Set-UID** menjalankan program yang dibuat oleh user sehingga muncul isinya dengan keterangan "Hacked!". Hal ini disebabkan oleh **Set-UID** yang dipanggil adalah bukan **PATH Absolut**, maka terjadi **PATH Attack** yang Dimana attacker dapat mengganti program yang dijalankan.

Task 7: The LD_PRELOAD Environment Variable.

A terminal window titled 'ekji@joy: ~/Downloads/Labsetup' showing a series of commands and their outputs. The user navigates to the 'Downloads/Labsetup' directory, creates a file 'mylib.c' with 'nano mylib.c', compiles it with 'gcc -fPIC -g -c mylib.c', creates a shared library 'libmylib.so.1.0.1' with 'gcc -shared -o libmylib.so.1.0.1 mylib.o -lc', creates a program 'myprog.c' with 'nano myprog.c', compiles it with 'gcc myprog.c -o myprog', sets the environment variable 'LD_PRELOAD=./libmylib.so.1.0.1' with 'export LD_PRELOAD=./libmylib.so.1.0.1', and runs the program with './myprog'. The output of the program is 'I am not sleeping!'.

```
ekji@joy: ~/Downloads/Labsetup
ekji@joy: ~/Downloads/Labsetup$ cd Downloads/Labsetup
ekji@joy: ~/Downloads/Labsetup$ nano mylib.c
ekji@joy: ~/Downloads/Labsetup$ gcc -fPIC -g -c mylib.c
ekji@joy: ~/Downloads/Labsetup$ gcc -shared -o libmylib.so.1.0.1 mylib.o -lc
ekji@joy: ~/Downloads/Labsetup$ nano myprog.c
ekji@joy: ~/Downloads/Labsetup$ gcc myprog.c -o myprog
ekji@joy: ~/Downloads/Labsetup$ export LD_PRELOAD=./libmylib.so.1.0.1
ekji@joy: ~/Downloads/Labsetup$ ./myprog
I am not sleeping!
ekji@joy: ~/Downloads/Labsetup$
```

Pada task ini terdapat **shared library** yang menggantikan fungsi tertentu, **library** ini dipaksa untuk memuat menggunakan environment variable **export LD_PRELOAD=./libmylib.so**, Ketika program dijalankan, dynamic linker akan memuat **library** tersebut terlebih dahulu sehingga fungsi asli digantikan oleh fungsi buatan **attacker** dan menampilkan pesan “**I am not sleeping!**”. Command ini memungkinkan library external dimuat sebelum library standar sehingga dapat mengubah perilaku program secara **dinamis**.

Task 8: The LD_PRELOAD Environment Variable and Set-UID Programs.

```
UTM File Edit Virtual Machine View Window Help
Activities Terminal
Mar 11 19:12
ekjl@joyy: ~/Downloads/Labsetup

ekjl@joyy:~/Downloads/Labsetup$ gcc catall.c -o catall
ekjl@joyy:~/Downloads/Labsetup$ sudo chown root catall
[sudo] password for ekjl:
ekjl@joyy:~/Downloads/Labsetup$ sudo chmod 4755 catall
ekjl@joyy:~/Downloads/Labsetup$ ./catall /etc/passwd

root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lpx:x:7:7:/var/ftp/pub:/usr/sbin/nologin
nail:x:8:8:nail:/var/nail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:MailList Manager:/var/list:/usr/sbin/nologin
ircd:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
_apt:x:100:65534:nonexistent:/usr/sbin/nologin
system-network:x:101:102:system Network Management,,,:/run/systemd:/usr/sbin/nologin
system-resolve:x:102:103:system Resolver,,,:/usr/sbin/nologin
messagebus:x:103:104:nonexistent:/usr/sbin/nologin
system-timesync:x:104:105:system Time Synchronization,,,:/run/systemd:/usr/sbin/nologin
pollinate:x:105:11:/var/cache/pollinate:/bin/false
syslog:x:106:111:/home/syslog:/usr/sbin/nologin
utdd:x:107:114:/run/utdd:/usr/sbin/nologin
tcpdump:x:108:115:nonexistent:/usr/sbin/nologin
tss:x:109:116:7m software stack,,,:/var/lib/tss:/bin/false
landscape:x:110:117:/var/lib/landscape:/usr/sbin/nologin
fwupd-refresh:x:111:118:fwupd-refresh user,,,:/run/systemd:/usr/sbin/nologin
ekit:x:1000:1000:xyz Fritz Firstwall:nonexistent:/bin/bash
lxd:x:999:100:/var/snap/lxd/common/lxd:/bin/false
rtkit:x:112:119:realtimekit,,,:/proc:/usr/sbin/nologin
dnsmasq:x:113:65534:dnsmasq,,,:/var/lib/misc:/usr/sbin/nologin
kernoops:x:114:65534:kernel Oops Tracking Daemon,,,:/usr/sbin/nologin
systemd-nova:x:115:122:systemd Nova Space Don Killer,,,:/run/systemd:/usr/sbin/nologin
whoopie:x:116:123:nonexistent:/bin/false
awahi-autopid:x:117:125:Awahi autopid daemon,,,:/var/lib/awahi-autopid:/usr/sbin/nologin
utbmux:x:118:46:utbmux daemon,,,:/var/lib/utbmux:/usr/sbin/nologin
nn-opensvnx:x:119:126:NetworkManager OpenVPN,,,:/var/lib/opensvnx/chroot:/usr/sbin/nologin
awahi:x:120:127:Awahi mDNS daemon,,,:/run/awahi-daemon:/usr/sbin/nologin
cps-pk-helper:x:121:128:user for cps-pk-helper service,,,:/home/cps-pk-helper:/usr/sbin/nologin
sssd:x:122:129:SSSD system user,,,:/var/lib/sss:/usr/sbin/nologin
speech-dispatcher:x:123:29:Speech Dispatcher,,,:/run/speech-dispatcher:/bin/false
samedx:124:131:/var/lib/samed:/usr/sbin/nologin
colord:x:125:132:colord colour management daemon,,,:/var/lib/colord:/usr/sbin/nologin
geoclue:x:126:133:/var/lib/geoclue:/usr/sbin/nologin
pulse:x:127:134:pulseaudio daemon,,,:/run/pulse:/usr/sbin/nologin
gnome-initial-setup:x:128:65534:/run/gnome-initial-setup:/bin/false
httpd:x:129:75:HTTP system user,,,:/run/httpd:/bin/false
gdm:x:130:136:Gnome Display Manager:/var/lib/gdm:/bin/false
seed:x:1001:1001:/home/seed:/bin/bash
telnetd:x:131:137:nonexistent:/usr/sbin/nologin
ekjl@joyy:~/Downloads/Labsetup$ ./catall "test; rm file"
/bin/cat: test: No such file or directory
ekjl@joyy:~/Downloads/Labsetup$ ./catall "test; rm /important/file"
/bin/cat: test: No such file or directory
ekjl@joyy:~/Downloads/Labsetup$ ./catall "test; rm /important/file"
/bin/cat: test: No such file or directory
ekjl@joyy:~/Downloads/Labsetup$ ./catall "test; rm /important/file"
rm: cannot remove '/important/file': No such file or directory
ekjl@joyy:~/Downloads/Labsetup$ ./catall "test; rm file"
/bin/cat: test: No such file or directory
ekjl@joyy:~/Downloads/Labsetup$ ./catall "test; rm file"
rm: cannot remove 'file': No such file or directory
ekjl@joyy:~/Downloads/Labsetup$
```

Pada Linux modern, **LD_PRELOAD** tidak diproses untuk program **Set-UID**, maka dari itu disini diujilah **LD_PRELOAD** apakah masih dapat digunakan pada program **Set-UID**, hal ini dilakukan untuk mencegah **attacker** mengganti fungsi penting pada program yang berjalan dengan privilege root. Hal ini dilakukan Linux modern untuk alasan keamanan, **dynamic linker** mengabaikan **LD_PRELOAD** pada program **Set-UID**, sehingga cara ini tidak dapat digunakan untuk **privilege escalation**.

Task 9: Privilege Escalation vis Set-UID.

The screenshot shows a terminal window titled "Terminal" with a dark background. The user is logged in as "ekjl@joyr" and is working in the directory "/Downloads/Labsetup". The terminal output shows the execution of several commands to install various system components:

```
ekjl@joyr:~/Downloads/Labsetup$ gcc catal.c -o catall
ekjl@joyr:~/Downloads/Labsetup$ sudo chown root catall
[sudo] password for ekjl:
ekjl@joyr:~/Downloads/Labsetup$ sudo chmod 755 catall
ekjl@joyr:~/Downloads/Labsetup$ ./catall /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
ucsp:x:10:10:ucsp:/usr/spool/ucsp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:36:36:Maillog List Manager:/var/list:/usr/sbin/nologin
ircd:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
gnats:x:41:41:Gnats Bug-Reporting System (admin)/var/lib/gnats:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
_apt:x:100:65534:./nonexistent:/usr/sbin/nologin
systemd-udevd:x:102:102:systemd Network Management,,,:/run/systemd:/usr/sbin/nologin
systemd-resolve:x:102:102:systemd Resolver,,,:/run/systemd:/usr/sbin/nologin
messagbus:x:103:104:./nonexistent:/usr/sbin/nologin
systemd-timesync:x:104:103:systemd Time Synchronization,,,:/run/systemd:/usr/sbin/nologin
pollinate:x:105:1:./var/cache/pollinate:/bin/false
syslog:x:106:113:./home/syslog:/usr/sbin/nologin
uidmap:x:107:145:./run/udev:/usr/sbin/nologin
tcpdump:x:108:115:./nonexistent:/usr/sbin/nologin
tssd:x:109:116:IBM software stack,,,:/var/lib/tssd:/bin/false
landscape:x:110:117:./var/lib/landscape:/usr/sbin/nologin
fwupd-refresh:x:111:118:fwupd-refresh user,,,:/run/systemd:/usr/sbin/nologin
etkls:x:100:300:xyz Fritz FirstStep:/home/ekjl/bin/bash
ldx:x:999:100:./var/snaps/ld/common/ld:/bin/false
rkkit:x:112:119:RealTimeKit,,,:/proc:/usr/sbin/nologin
dnsmasq:x:113:65534:dnsmasq,,,:/var/lib/ndisc:/usr/sbin/nologin
kerneloops:x:114:65534:Kernel Oops Tracking Daemon,,,:/usr/sbin/nologin
systemd-core:x:115:122:systemd Uppspace OOM Killer,,,:/run/systemd:/usr/sbin/nologin
mtools:x:116:121:./nonexistent:/bin/false
avahi-autoipd:x:117:125:Avahi autoipd daemon,,,:/var/lib/avahi-autoipd:/usr/sbin/nologin
usbmuxd:x:118:10:usbmuxd daemon,,,:/var/lib/usbmuxd:/usr/sbin/nologin
nsg-openvpn:x:119:116:NetworkManager OpenVPN,,,:/var/lib/openvpn/chroot:/usr/sbin/nologin
nsd:x:120:127:Avahi mDNS daemon,,,:/run/avahi-daemon:/usr/sbin/nologin
cpio-pk-helper:x:121:128:user for cpio pk-helper service,,:/home/cups-pk-helper:/usr/sbin/nologin
sshd:x:122:129:SSHD system user,,,:/var/lib/ssh:/usr/sbin/nologin
speech-dispatcher:x:123:29:Speech Dispatcher,,,:/run/speech-dispatcher:/bin/false
scitools:x:124:131:./var/lib/faneddy:/usr/sbin/nologin
colorctl:x:125:132:colorctl colour management daemon,,,:/var/lib/colorctl:/usr/sbin/nologin
gocueuler:x:126:133:./var/lib/gocueuler:/usr/sbin/nologin
polite:x:127:134:PulseAudio daemon,,,:/run/polite:/usr/sbin/nologin
gnome-initial-setup:x:128:65534:/run/gnome-initial-setup:/bin/false
holips:x:129:79:HolIPs system user,,,:/run/holips:/bin/false
ghm-x:130:136:Goome Display Manager:/var/lib/ghm:/bin/false
seed:x:1001:10001:/home/seed:/bin/bash
telettyd:x:131:137:./nonexistent:/usr/sbin/nologin
ekjl@joyr:~/Downloads/Labsetup$ ./catall test; rm file'
/bin/cat: test: No such file or directory
rm: cannot remove 'file': No such file or directory
ekjl@joyr:~/Downloads/Labsetup$ ./catall test; rm !important/file'
/bin/cat: test: No such file or directory
rm: cannot remove '!important/file': No such file or directory
ekjl@joyr:~/Downloads/Labsetup$ rm !important/file
rm: cannot remove '!important/file': No such file or directory
ekjl@joyr:~/Downloads/Labsetup$ ./catall test; rm file'
/bin/cat: test: No such file or directory
rm: cannot remove 'file': No such file or directory
```

Pada task ini dibuat program **Set-UID** yang menjalankan **shell** dengan permission:

sudo chown root root

```
sudo chmod 4755 root
```

Ketika dijalankan oleh user biasa, output menunjukkan:

Real UID: 1000

Effective UID: 0

Yang dimana berarti program berjalan pada **privilege root**, walaupun **user** yang menjalankan bukan **root**. Kemudian program membuka **shell** sehingga **user** mendapatkan akses root shell **uid=0(root)**, hal ini menunjukkan bahwa **Set-UID** dapat membuat **privilege escalation** jika program tidak dirancang dengan aman.